Assignment 5

CS 6117 (Fall 2025)

Exercise 1. The indexed type system System S supports universal quantification over indices/sizes. In this exercise, we will add types that *existentially* quantify over indices numbers to System S. For simplicity, we suppose the index and type signature Sg^{ty} has just a single index kind κ (although supporting multiple index kinds should be entirely analogous). We want the following type, along with a well-formedness rule:

$$\Phi ::= \cdots \mid \exists X : \kappa.\, \Phi \qquad\qquad \frac{\Delta, X \text{ is } \kappa \vdash \Phi \text{ is } Type}{\Delta \vdash \exists X : \kappa.\, \Phi \text{ is } Type}$$

We also want two new terms:

$$M ::= \cdots \mid \mathsf{pack}\ (I, M) \mid \mathsf{open}\ (X, x) = M\ \mathsf{in}\ M.$$

Just like in the notes, I is an *index expression*; the grammar is not important. We want the following two typing rules:

$$\frac{\Delta \vdash I \text{ is } \kappa \qquad \Delta, X \text{ is } \kappa \vdash \Psi \text{ is } Type \qquad \Delta \mid \Gamma \vdash M : \Psi[I/X]}{\Delta \mid \Gamma \vdash \mathsf{pack}\ (I, M) : \exists X : \kappa.\, \Psi}$$

$$\frac{\Delta \mid \Gamma \vdash M : \exists X : \kappa.\, \Phi \qquad \Delta, X \text{ is } \kappa \mid \Gamma, x : \Phi \vdash N : \Psi}{\Delta \mid \Gamma \vdash \mathsf{open}\ (X, x) = M\ \mathsf{in}\ N : \Psi}$$

Intuitively, pack makes a program of existential type by packaging a index expression I and a term, while open uses a program M of existential type by binding the contents of the package to an index and term variable, which can be used in the body N. We follow the CfT convention that premises and conclusions are assumed to be well-formed; in particular, the index variable X never appears free in the types of the term context Γ. Finally, we want the following equations:

$$\frac{\Delta \vdash I \text{ is } \kappa \qquad \Delta \mid \Gamma \vdash M : \Theta[I/X] \qquad \Delta, X \text{ is } \kappa \mid \Gamma, x : \Theta \vdash N : \Psi}{\Delta \mid \Gamma \vdash (\mathsf{open}\ (X, x) = \mathsf{pack}\ (I, M)\ \mathsf{in}\ N) = N[I/X, M/x] : \Psi}\ \beta\text{-EXISTS}$$

$$\frac{\Delta \mid \Gamma \vdash M : \exists X : \kappa.\, \Phi \qquad \Delta \mid \Gamma, z : \exists X.\, \Phi \vdash N : \Psi}{\Delta \mid \Gamma \vdash (\mathsf{open}\ (X, x) = M\ \mathsf{in}\ N[(\mathsf{pack}\ (X, x))/z]) = N[M/z] : \Psi}\ \eta\text{-EXISTS}$$

Intuitively, the β rule says that if we pack a index and a term, and then immediately open the package, the result is the same as just substituting the index and term into the body. The substitution into N first substitutes I, then substitutes M. The η rule says that if we open a package and then immediately pack the variables for use in some term, the result is the same as just substituting the package into the term.

Let's add these types and terms, along with the typing rules and equations β-EXISTS and η-EXISTS, to the language. We will work through how to adapt the semantics of System S presented in the notes to handle this extension. Let $\mathbb{C} : \mathbf{Set}^{op} \to \mathbf{CCCat}$ be any indexed category. For every pair of base objects $A, B \in Ob(\mathbf{Set})$, there is a projection morphism $\pi_{A,B} : A \times B \to A$ and a reindexing functor $\pi^* : \mathbb{C}(A) \to \mathbb{C}(A \times B)$. We've seen that we can model model universal quantification using a *right* adjoint $\forall_{A,B}$ to the reindexing functor π^*.

Throughout this problem, suppose instead that there is a *left* adjoint $\exists_{A,B} : \mathbb{C}(A \times B) \to \mathbb{C}(A)$, i.e.,

$$\overline{(-)} : \mathbb{C}(A)(\exists(\psi), \phi) \rightleftarrows \mathbb{C}(A \times B)(\psi, \pi^*(\phi)) : \widehat{(-)}$$

As the notation suggests, we can use this functor to interpret existential types.

1. Define the categorical semantics of existential types as an object in a fiber category.

2. Define the semantics of the existential term pack. [Hint: take a look at how we defined the semantics of index application; you may want to look at the units/counits of the adjunction and see if there is a useful morphism that you can use.]

3. Define the semantics of the existential term open. You can assume that for any object X in $\mathbb{C}(A \times B)$ and object Y in $\mathbb{C}(A)$, you have a morphism in the category $\mathbb{C}(A)$ of the form:

   $$\rho : \exists_{A,B}(X) \times Y \to \exists_{A,B}(X \times \pi^*_{A,B}(Y))$$

   Intuitively, this morphism pulls an object that doesn't mention the existential variable into the existential.

   In categorical semantics, the map ρ is often an isomorphism: this condition about an adjoint pair and categorical products is called the *Frobenius condition*. A sufficient condition is for the right-adjoint to be a cartesian closed functor—in our setting, the right-adjoint is re-indexing $\pi^*_{A,B}$, which is cartesian closed. Since $\pi^*_{A,B}$ *strictly* preserves exponentials, it is known that ρ can be explicitly defined as follows:

   $$\rho \triangleq \widehat{\lambda}([\lambda(\overline{id_{\exists_{A,B}(X \times \pi^*_{A,B}(Y))}})]^\wedge)$$

   where λ curries a function, and $\widehat{\lambda}$ uncurries a function. You can verify that this morphism has the right type.

4. (**Sparkly star**) Prove that your categorical semantics satisfies the equation β-EXISTS. (The equation η-EXISTS is also supposed to be sound, but I haven't worked through the details myself because ρ is a bit ugly.)

5. (**Sparkly star**) Let $\mathbf{D}$ be any Cartesian closed category with small (set-indexed) coproducts, and consider the concrete example of an indexed category $\mathbb{C}$ where $\mathbb{C}(B) = \mathbf{D}^B$, described in the notes. Define the left-adjoint to weakening, and prove that it is an adjoint.

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1. The interpretation of types is in the fiber categories $C([[\Delta]])$, where $[[\Delta]]$ is the object in Set interpreting the index context Δ. For $\Delta \vdash \exists X : \kappa.\Phi$ $\mathsf{is\ Type}$, with $[\Delta]$ the set interpreting the context,

$$[\Delta \vdash \exists X : \kappa.\Phi] = \exists_{[\Delta],[\kappa]}\left([\Delta, X : \kappa \vdash \Phi]\right) \in \mathsf{Ob}\big(C([\Delta])\big).$$

   This applies the existential quantifier functor to the body interpreted in the extended fiber $C([\Delta]\times[\kappa])$, capturing dependency on the hidden index.

2. Consider $[\![\Delta]\!] = A$, $[\![\kappa]\!] = B$, $[\![I]\!] : A \to B$, and $[\![\Psi]\!] \in \mathsf{Ob}(C(A \times B))$. The substitution morphism $\sigma = \langle \mathsf{id}_A, [\![I]\!]\rangle : A \to A \times B$ induces $[\![\Psi[I/X]]\!] = \sigma^*([\![\Psi]\!])$ in $C(A)$. Given $[\![M]\!] : [\![\Gamma]\!] \to \sigma^*([\![\Psi]\!])$ in $C(A)$, and letting η denote the unit of the adjunction $\exists_{A,B} \dashv \pi^*_{A,B}$, the semantics is $[\![\mathsf{pack}(I, M)]\!] = \sigma^*(\eta_{[\![\Psi]\!]}) \circ [\![M]\!] : [\![\Gamma]\!] \to \exists_{A,B}([\![\Psi]\!])$.

3. Let $[\![M]\!] : [\![\Gamma]\!] \to \exists_{A,B}([\![\Phi]\!])$ in $C(A)$, and $[\![N]\!] : \pi^*_{A,B}([\![\Gamma]\!]) \times [\![\Phi]\!] \to \pi^*_{A,B}([\![\Psi]\!])$ in $C(A \times B)$. Using the provided morphism $\rho : \exists_{A,B}(X) \times Y \to \exists_{A,B}(X \times \pi^*_{A,B}(Y))$ with $X = [\![\Phi]\!]$ and $Y = [\![\Gamma]\!]$, along with the symmetry isomorphism $\tau : Y \times \exists_{A,B}(X) \to \exists_{A,B}(X) \times Y$ in the CCC $C(A)$, and letting ε be the counit of the adjunction, the semantics is $[\![\mathsf{open}(X, x) = M \ \mathsf{in}\ N]\!] = \varepsilon_{[\![\Psi]\!]} \circ \exists_{A,B}([\![N]\!]) \circ \rho \circ \tau \circ \langle \mathsf{id}_{[\![\Gamma]\!]}, [\![M]\!]\rangle$. Here, τ adjusts the product order, ρ internalizes the context under the existential, $\exists_{A,B}([\![N]\!])$ applies the body, and ε eliminates the quantifier.

4. We verify $\mathsf{open}(X, x) = \mathsf{pack}(I, M)\ \mathsf{in}\ N = N[I/X, M/x]$. The right-hand side interprets as the substitution $[\![N[I/X, M/x]]\!]$, which reindexes along σ and composes with $[\![M]\!]$. For the left-hand side, substitute the semantics of pack: $[\![\mathsf{open}]\!] = \varepsilon_{[\![\Psi]\!]} \circ \exists_{A,B}([\![N]\!]) \circ \rho \circ \tau \circ \langle \mathrm{id}_{[\![\Gamma]\!]}, \sigma^*(\eta_{[\![\Theta]\!]}) \circ [\![M]\!]\rangle$, where $\Theta = \Phi$. By naturality of η and ε, the Frobenius condition (making ρ interact coherently with products and reindexing), and the triangular identities of the adjunction ($\varepsilon \circ \exists(\eta) = \mathrm{id}$ and $\pi^*(\varepsilon) \circ \eta = \mathrm{id}$), the composition simplifies: the unit and counit cancel through the existential, yielding precisely the substituted body. The Beck-Chevalley condition ensures compatibility with the substitution σ, confirming equality.

5. Let D be a CCC with small coproducts. Define $C(B) = D^B$ (functors from the discrete category on B to D), with reindexing $u^*(G) = G \circ u$ for

$u : A' \to A$. Then $\pi^*_{A,B}(F)(a,b) = F(a)$. The left adjoint is $\exists_{A,B}(G)(a) = \coprod_{b \in B} G(a,b)$. The adjunction holds since

$$\begin{aligned} C(A)(\exists_{A,B}(G), F) &= \prod_a D\left(\coprod_b G(a,b), F(a)\right) \\ &\cong \prod_a \prod_b D(G(a,b), F(a)) \\ &= \prod_{(a,b)} D(G(a,b), F(a)) \\ &= C(A \times B)(G, \pi^*_{A,B}(F)) \end{aligned}$$

by the product-coproduct duality in D.

Exercise 2. In this exercise, we establish a few properties about fibrations.

1. Suppose that $p : \mathbf{E} \to \mathbf{C}$ is a fibration. Show that Cartesian liftings are unique up to isomorphism: if $f : A \to Z$ and $f' : B \to Z$ are both Cartesian morphisms in $\mathbf{E}$ over the same map u in $\mathbf{C}$, then there is a unique vertical morphism $\varphi : A \to B$ with $f = f' \circ \varphi$ and φ is an isomorphism.

2. Suppose that $p : \mathbf{E} \to \mathbf{D}$ and $q : \mathbf{D} \to \mathbf{C}$ be two fibrations. Show that the composite $qp : \mathbf{E} \to \mathbf{C}$ is also a fibration. In this way, fibrations can smoothly extend to model languages with multiple levels of dependency. (If you are curious, think about how you would state this theorem for indexed categories.)

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1. Let $p : E \to C$ be a fibration. For $u : X \to Y$ in C and Z over Y, suppose $f : A \to Z$, $f' : B \to Z$ are Cartesian lifts of u (so $p(A) = X = p(B)$, $p(f) = u = p(f')$). Apply f''s Cartesianness to $f : A \to Z$: Since $p(f) = u = u \circ \mathsf{id}_X$, there exists unique $\phi : A \to B$ over id_X (vertical) with $f = f' \circ \phi$. Symmetrically, f's Cartesianness yields unique vertical $\psi : B \to A$ with $f' = f \circ \psi$. Composition gives $\psi \circ \phi = \mathrm{id}_A$ (uniqueness: id_A also satisfies $f = f \circ \mathsf{id}_A$); similarly $\phi \circ \psi = \mathrm{id}_B$. Thus, ϕ is an isomorphism.

2. Let $p : E \to D$, $q : D \to C$ be fibrations; show $q \circ p : E \to C$. For $u : X \to Y$ in C and $Z \in E$ with $(qp)(Z) = Y$: First, lift u via q at $p(Z)$ (over Y): Cartesian $g : M \to p(Z)$ in D with $q(g) = u$, $q(M) = X$. Then, lift g via p at Z: Cartesian $h : W \to Z$ in E with $p(h) = g$, $p(W) = M$. Set $f = h$; then $(qp)(f) = q(g) = u$. For Cartesianness: Given $k : K \to Z$ in E with $(qp)(k) = u \circ r$, $r : (qp)(K) \to X$. Since q's Cartesian: $p(k) : p(K) \to p(Z)$ in D, $q(p(k)) = u \circ r = q(g) \circ r$, so

unique $s : p(K) \to M$ over r (for q) with $p(k) = g \circ s$. Since p's Cartesian: $k : K \to Z$, $p(k) = g \circ s = p(h) \circ s$, so unique $t : K \to W$ over s (for p) with $k = h \circ t$. Then $(qp)(t) = q(p(t)) = q(s) = r$. Uniqueness follows from the chained uniqueness of s and t.

Exercise 3 (Category of the Week). Let $\mathbf{C}$ be any category. A *span* in the category $\mathbf{C}$ is a diagram of the form $A \leftarrow X \to B$, i.e., a triple of objects (A, X, B) along with two morphisms $p_A : X \to A$ and $p_B : X \to B$. When $\mathbf{C} = \mathbf{Set}$, any binary relation $R \subseteq A \times B$ can be encoded as a span $A \leftarrow R \to B$, and vice versa. But spans make sense in any category $\mathbf{C}$.

1. A *span morphism* from $A \leftarrow X \to B$ to $C \leftarrow Y \to D$ consists of three morphisms in $\mathbf{C}$, namely $f : A \to C$, $g : B \to D$, and $\varphi : X \to Y$, making the evident diagrams commute. Spell out these conditions, and convince yourself that spans and span morphisms form a category.

   When $\mathbf{C} = \mathbf{Set}$, a span morphism encodes a *relation preserving* pair of maps, i.e., $f : A \to C$ and $g : B \to D$ such that pairs of elements in the relation X map to pairs of elements in the relation Y. But a span morphism encodes more information: the intermediate map φ maps a witness that (a, b) are related to a witness that (c, d) are related. Thus, one way to think of spans is as *proof relevant* relations: not only do they encode which pairs of elements are related, but they also encode *how* a pair of elements are related, and a given pair of elements might be related in more than one way.

2. Spans can also be *composed*: given a span $A \leftarrow X \to B$ and $B \leftarrow Y \to C$ in a category $\mathbf{C}$ with pullbacks, show how to define a span $A \leftarrow Z \to C$ for some object Z.

3. Given three spans $A \leftarrow X \to B$, $B \leftarrow Y \to C$, and $C \leftarrow Z \to D$, we can compose these spans in two different ways to build a span from A to D: we either compose the first pair first, or the second pair first. Show that the two resulting spans are isomorphic (in the category of spans and span morphisms).

4. Show that the span $A \leftarrow A \to A$, where the morphisms are both identity, behaves like the identity under span composition: namely, composing with $A \leftarrow X \to B$ on the right gives an isomorphic span, and similarly composing with a span on the left.

If we try to build a category $\mathrm{Span}(\mathbf{C})$ where objects are objects from $\mathbf{C}$ and morphisms from A to B are spans $A \leftarrow X \to B$ under span composition, the previous exercises show that span composition does not satisfy the identity and associativity laws exactly (i.e., *strictly*), as is required in a standard category. But the identity and associativity laws are satisfied up to isomorphism (i.e.,

weakly) for a notion of morphism between morphisms. These isomorphisms are called the *unitors* and the *associator*; in this example, these maps are span (iso)morphisms.

This makes Span(**C**) a leading example of a *bicategory*, a category where there are both (1-) morphisms, and (2-) morphisms between (1-) morphisms. We have already seen another example: the category **Cat** of categories, functors, and natural transformations, but since composition of functors satisfies the identity and associativity laws exactly, **C** is an example of a *strict* 2-category.

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1. Objects are those of C. A morphism from A to B is a span $A \xleftarrow{p_A} X \xrightarrow{p_B} B$. A span morphism from $A \xleftarrow{p_A} X \xrightarrow{p_B} B$ to $C \xleftarrow{p_C} Y \xrightarrow{p_D} D$ comprises $f : A \to C$, $g : B \to D$, and $\phi : X \to Y$ such that $p_C \circ \phi = f \circ p_A$ and $p_D \circ \phi = g \circ p_B$. Composition is componentwise, with identities $(\mathsf{id}_A, \mathsf{id}_B, \mathsf{id}_X)$. Associativity and units follow from C, forming a category.

2. For spans $A \xleftarrow{p} X \xrightarrow{q} B$ and $B \xleftarrow{r} Y \xrightarrow{s} C$, form the pullback Z of q and r, with projections $\pi_X : Z \to X$ and $\pi_Y : Z \to Y$. The composite is $A \xleftarrow{p \circ \pi_X} Z \xrightarrow{s \circ \pi_Y} C$.

3. For spans $A \leftarrow X \to B$, $B \leftarrow Y \to C$, $C \leftarrow W \to D$: Composites $(X \circ Y) \circ W$ and $X \circ (Y \circ W)$ yield pullbacks $(X \times_B Y) \times_C W \cong X \times_B (Y \times_C W)$ (associativity of pullbacks up to canonical isomorphism in C). This iso $\alpha : (X \times_B Y) \times_C W \to X \times_B (Y \times_C W)$ forms a span morphism with $\mathrm{id}_A, \mathrm{id}_D$ (diagrams commute by projection properties).

4. The span $A \xleftarrow{\mathrm{id}} A \xrightarrow{\mathrm{id}} A$ is unit: Composing right with $A \leftarrow X \to B$ pulls back $A \times_A X \cong X$ (universal: $\mathsf{id} \circ p_A = p_A = p_A \circ \mathsf{id}_X$). The iso $\beta : A \times_A X \to X$ (the projection) forms a span morphism with $\mathsf{id}_A, \mathsf{id}_B$. Similarly for left composition.
